PROJECT AND TEAM INFORMATION

Project Title

EduParse: An Education Parser Tool

Student / Team Information

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PROPOSAL DESCRIPTION

Motivation

Nowadays, the role of programming language and compiler is very much important, especially for the computer science student. To understand the parsing concept for different parser (such as SLR, CLR, LALR, etc.) is difficult for the students both theoretically and concept wise. In traditional classroom, students do not completely visualize the mechanism of the parsers which make it difficult for them to understand the actual working of the parser.

The main motivation of this project is to develop a system which shows the step-by-step process of the parsers in visualize manner. This tool takes grammar as input, generate the parsing table and show parsing steps clearly through which our learning became easier. Along with this the importance of this project is that the compiler design is a fundamental subject so strong foundation helps us to build compiler, interpreter and other technologies in future.

The aim of this project is to create educational tool through which we can convert the theoretical knowledge into practical concepts and to understand parsing concepts and grammar error in a better way.

State of the Art / Current solution

Currently, parsing concepts are taught mostly through online concept and notebooks. Some online compilers are also available for this but those are little bit complicated and difficult to understand as a beginner. Many existing Compiler give direct output without internal working due to which the students do not understand the shift-reduce concept and parsing table creation.

To fulfil these requirements, we have created a simple user-friendly parser tool which act as educational tool for the students to visualize the concepts.

Project Goals and Milestones

The primary goal of this project is to develop efficient and easy-to-use parsers which analyzes the grammar and generate the correct parsing output. The objective of the system is to automate the parsing process and give real-time understanding to the students.

In initial milestone, we develop the module to analyze the grammar rules and then in next step we generate the item sets and construct the parsing table. After that parser implementation will occur which validate the string either accepted or rejected.

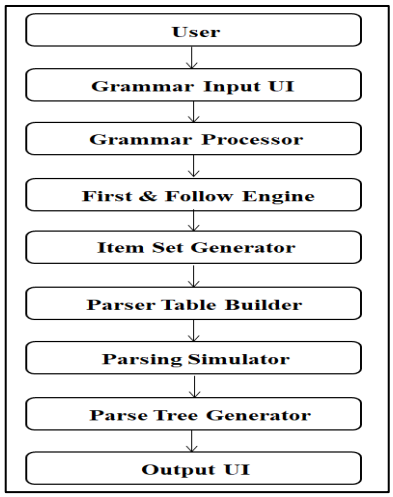
Overall, the goal is to develop a parser tool which enhance our learning and make the compiler design concepts practical.

Project Approach

To develop this project, we follow a structured approach. First, we make a grammar module which read the sets of input, then compute the First and follow which is essential for parsing. In next step we calculate the sets of items and then create the parsing table in which shift, reduce, accept and error states are present.

Parsing algorithm will analyze the input string and used the stack-based mechanism to perform the parsing. At the parsing time we also make a parsing tree which include shift and reduce operation. Testing will be done from multiple grammar through which are accuracy is better.

System Architecture



This system architecture represents the working of the parser at a high-level. Through this diagram we show the clear data flow between the system components. The process of system architecture will start from user input where user enter the grammar which will stored in the input buffer temporarily. After that, the data reached to the parser, which analyze the grammar with the help of parser algorithm. At the time of parsing, the parser will make use of stack to maintain the states and symbols to maintain the parsing table and to decide the next step reduce, shift and accept. Then, the parsing process will complete and system will generate the output that the string is accepted or not.

In this way, it will make the parsing systematic, organized and well-structured.

Project Outcome / Deliverables

The outcome of this project is a fully working parser tool which analyze the grammar and demonstrate the parsing process. Deliverables include the source code, documentation and user guide.

It is useful for students because it makes the complex parsing process easier and explain it in a proper format. With the help of this user will understand the grammar logic and parsing process.

This is useful for the parser concept to explain it through visualization to make the concept and everything much easier.

This includes the input module, table and tree creation module and output module.

Assumptions

We are assuming that the user will understand the basic context-free grammar and compiler concepts. System will expect a valid grammar input and follow the standard parsing rule. This project is developed as an educational tool that's why the focus is on clarity and correctness rather than performance.

References

1. *GeeksforGeeks – Compiler design section - https://www.geeksforgeeks.org/compiler-design/parser/*
2. *TutorialPoint – LR parsing - https://www.tutorialspoint.com/what-is-types-of-lr-parser-in-compiler-design*
3. *Wikipedia – parsing - https://en.wikipedia.org/wiki/LR_parser*